BUILDING AN EXCEL DASHBOARD

Step-by-step guide using the FIFA Player Dataset

From raw data to KPIs, PivotTables, charts and an interactive report

Dataset loaded: 18,207 player records | 89 columns

Learning objectives

By the end of this activity, learners will be able to:

- Inspect and clean a real-world player dataset in Excel.
- Create calculated fields and summary measures for player analytics.
- Build PivotTables and PivotCharts to answer business questions.
- Arrange KPI cards, charts and filters into a readable dashboard.
- Explain insights without confusing correlation, potential and current performance.

1. Understand the dataset

Start by checking the structure before building visuals.

PLAYER RECORDS	COLUMNS	MISSING CELLS	MISSING-CELL RATE
18,207	89	76,984	4.8%

Useful fields found: Age, Overall, Potential, Club, Nationality, Position, Value, Wage.

In Excel: open the CSV, convert the data range to a Table with Ctrl + T, and confirm that each column has one clear header.

Keep a raw-data copy before cleaning; do not delete records just because a field is missing.

2. Prepare and clean the data

Use consistent data types before creating summaries.

Check blank cells: use Filter → (Blanks), or Home → Find & Select → Go To Special → Blanks.

Check duplicate records: Data → Remove Duplicates, but only after deciding which fields uniquely identify a player.

Confirm Age, Overall and Potential are numeric; use Data → Text to Columns or VALUE() if numbers are stored as text.

Standardize category labels such as Club, Nationality and Position; watch for extra spaces.

For Value and Wage fields containing K/M suffixes, create numeric helper columns before summing or averaging.

3. Add useful helper columns

Example formulas (assume the Excel Table is named Players).

Age Group: =IF([@Age]<21,"Under 21",IF([@Age]<=25,"21–25",IF([@Age]<=30,"26–30","Over 30")))

Rating Band: =IF([@Overall]>=85,"Elite",IF([@Overall]>=75,"High",IF([@Overall]>=65,"Mid","Developing")))

Potential Gap: =[@Potential]-[@Overall]

Check formula outputs for blank or non-numeric values before using them in PivotTables.

If your headers differ, replace the structured references with your actual column names.

4. Select dashboard KPIs

Use a small number of metrics that answer clear questions.

TOTAL PLAYERS	AVERAGE AGE	AVERAGE OVERALL	AVERAGE POTENTIAL
18,207	25.1	66.2	71.3

Excel formulas: =COUNTA(Players[Name]) for player count; =AVERAGE(Players[Age]) for average age.

Use =AVERAGE(Players[Overall]) and =AVERAGE(Players[Potential]) for rating summaries.

KPI definitions should be visible: overall rating is current rating, while potential is an estimate of future development.

5. Build a PivotTable

Example: compare player counts by club.

Click any cell in the Players table.

Choose Insert → PivotTable → From Table/Range → New Worksheet.

Drag Club to Rows and Name (or a unique player ID) to Values.

In Value Field Settings, choose Count rather than Sum.

Sort the player count from Largest to Smallest; apply a Top 10 filter if useful.

Repeat with Nationality, Position, Age Group and Rating Band.

ROWS → Club
VALUES → Count of players
FILTERS → Age Group / Nationality

6. Create PivotCharts

Choose the chart type to match the question.

Bar chart: top 10 clubs by number of players.

Bar chart: top nationalities by player count.

Column chart: player count by age group or rating band.

Horizontal bar chart: average Overall by position (if position labels are clean).

Avoid pie charts with many categories; use a bar chart when labels need comparison.

7. Arrange the dashboard

Keep the most important information at the top.

FIFA PLAYER ANALYTICS

Players	Avg Age	Avg Rating	Avg Potential
18K+	25.1	66.2	71.3

Top 10 Clubs by Player Count

Insert chart here

Players by Position

Insert chart here

Age Group Distribution

Insert chart here

Overall Rating Bands

Insert chart here

Place slicers along the right or top edge; keep colors, font sizes and number formats consistent.

8. Add interactive filters

Slicers let users explore the same dashboard from different angles.

Select a PivotTable → PivotTable Analyze → Insert Slicer.

Choose fields such as Club, Nationality, Preferred Foot, Age Group or Position.

To connect a slicer to multiple PivotTables: select slicer → Report Connections (or PivotTable Connections) → tick each relevant PivotTable.

Test every slicer. If a chart does not change, check its connection to the PivotTable.

For dates, use a Timeline only if the dataset contains a meaningful date field.

9. Dashboard design checklist

A dashboard should be readable before it is decorative.

Use a clear title, reporting context and concise KPI labels.

Use 2–3 coordinated colors; reserve a contrasting color for emphasis.

Use consistent number formats and avoid unnecessary decimals.

Sort bars by value, label axes, and show units where relevant.

Remove gridlines and chart borders when they do not help interpretation.

Keep raw data and PivotTables on separate worksheets; name sheets clearly.

10. Turn charts into insights

Describe what the dashboard shows, then explain what to investigate next.

Example: Which clubs have the largest player counts? Does this reflect roster size or dataset coverage?

Example: Which age group has the most players? How does average Overall vary across age groups?

Example: Which positions have a higher average rating? Check whether position groups have comparable sample sizes.

Example: Where is the gap between Potential and Overall largest? Treat this as a descriptive indicator, not a guaranteed development outcome.

Always distinguish observations from recommendations; validate surprising patterns against the source data.

Classroom activity: build your dashboard

Suggested time: 45–60 minutes.

Task 1 (10 min): format the dataset as a Table and review missing values.

Task 2 (10 min): create Age Group, Rating Band and Potential Gap helper columns.

Task 3 (15 min): create PivotTables for Club, Nationality, Position and Age Group.

Task 4 (10 min): create at least three charts and four KPI cards.

Task 5 (5–15 min): add slicers, format the layout and write three evidence-based observations.

Common problems and fixes

Use this checklist when a PivotTable or chart looks wrong.

Numbers appear as Count instead of Average: convert the source column to numeric, refresh, then change Value Field Settings.

Slicer does not filter a chart: check Report Connections and confirm the chart is based on the connected PivotTable.

New rows do not appear: use an Excel Table as the source and choose PivotTable Analyze → Refresh.

Duplicate player names: use a unique player ID if available for counting records.

Value/Wage totals look incorrect: convert K/M text to numeric values and verify units before summing.

FINAL DELIVERABLE

One-page FIFA Player Analytics Dashboard

4 KPI cards: player count, average age, average Overall, average Potential.

At least 4 charts: club count, nationality count, position distribution, age/rating distribution.

At least 3 slicers connected to the relevant PivotTables.

A clean layout with consistent formatting and 3 written insights.

Save as .xlsx and test filters before sharing.

Questions? Use the dashboard to answer a business question—not just to display charts.